V01.05 – Mirror Mathematics II

Minimal Observer Thresholds

From Reliability Variables to Recursive Observer Predicates

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Abstract

Mirror Mathematics I introduced a constraint calculus for recursive observerhood. It defined constraint regimes, viability functionals, bounded modelling, self-models, reliability variables and salience gain. The present paper develops the next formal step: a minimal observer threshold. The central claim is not that all self-referential systems are observers, nor that all predictive systems are observers. Rather, a system crosses the minimal Mirror observer threshold only when self-model reliability is represented, action-guiding, cost-sensitive and positively coupled to viability under perturbation.

Formally, observer-relevance is tied to a salience functional $\Sigma_L(R_L, \rho)$ measuring the expected viability benefit of using a self-reliability variable and its associated repair or recalibration policy, net of representational and action costs. The paper proves several necessary conditions: self-reference alone is insufficient; passive reliability is insufficient; reliability that does not change policy cannot be observer-salient under positive cost; and recursive reliability is justified only when first-order reliability itself is vulnerable to corruption.

The computational role of Mirror Observerhood Labs I–V is scaffolding, not proof of consciousness or artificial general intelligence. The now-published Lab sequence illustrates the formal threshold: reliability helps under self-relevant perturbation when repair is possible and cost-justified, but becomes neutral or harmful under low perturbation, excessive repair cost, channel mismatch or unnecessary meta-diagnosis. This paper positions the threshold layer of the Mathematics arc between the general constraint calculus and the later treatment of identity continuity, memory governance and physical support.

Keywords: Mirror Programme; Mirror Mathematics; observerhood; recursive observerhood; minimal observer threshold; self-models; reliability; viability; recursive modelling; salience; metacognition; active inference; cybernetics; artificial agents; constraint calculus.

Contents

1 Roadmap position

The Mirror Programme is organised around a shared formal spine: constraint, model, reliability, repair, memory, viability, continuity and scale [1]. Volume I applies that spine to observerhood and treats observerhood as a derived organisational condition rather than a primitive subject, witness or point of view [2].

Within Volume I, the Theory arc derives and positions recursive observerhood. Mirror Theory I develops the minimal computational derivation; Mirror Theory II examines the philosophical consequences for identity, reality and meaning; and Mirror Theory III situates the framework among neighbouring theories of cognition, consciousness and artificial intelligence [3, 4, 5]. Mirror Mathematics I then supplies the first mathematical layer: a constraint calculus for bounded local modelling, self-modelling, reliability and salience [6].

The present paper is the second Mathematics paper. Its task is narrower than the Theory arc and more specific than the general calculus. It asks when a reliability variable becomes observer-relevant rather than merely predictive, decorative or self-referential. It therefore turns the broad grammar of Mirror Mathematics I into a minimal observer predicate, informed by the five published computational Observerhood Labs, for later neuro-symbolic, memory-governance and physical-support work [7, 8, 9, 10, 11].

2 Core claim

The core claim is deliberately limited:

> A local subsystem crosses the minimal Mirror observer threshold only when it maintains a self-model whose reliability is represented, action-guiding, cost-sensitive and positively coupled to viability under perturbation.

The claim is not that every self-referential system is an observer. It is not that every system with a world-model is an observer. It is not that every language model becomes an observer when it describes itself. The threshold is functional and regime-relative. It depends on the coupling among self-model error, reliability estimation, repair policy, cost and expected viability.

3 Scope and negative commitments

This paper uses the word "observer" in a technical and minimal sense. The term denotes an organised constraint regime satisfying a threshold predicate, not a full theory of phenomenal consciousness.

3.1 Not a consciousness proof

The minimal Mirror observer predicate is not a claim that the relevant system has subjective experience, moral status or personhood. It may identify structural preconditions relevant to later theories of consciousness, but it does not by itself solve the hard problem of consciousness.

3.2 Not an artificial general intelligence claim

The computational labs discussed below are toy scaffolds. They test threshold logic under controlled perturbation. They do not demonstrate artificial general intelligence, biological equivalence or human-like subjectivity.

3.3 Not self-reference alone

A self-referential sentence, a program that prints its own source code, or a database containing metadata about itself does not satisfy the predicate merely by referring to itself. Observer-relevance requires viability-coupled self-model reliability.

3.4 Not monotonic recursion

The framework does not say that more self-monitoring, more recursion or more metacognition is always better. Recursive reliability becomes useful only where lower-order reliability can fail in viability-relevant ways and where the cost of diagnosis or repair is lower than the expected avoided loss.

4 Introduction

Mirror Theory begins from a deliberately modest premise: do not assume observers at the foundation. Instead, ask how observer-like organisation can arise within a constraint-governed system. Earlier papers in the programme formulated observerhood in terms of viability, bounded modelling, compression, environmental prediction, self-modelling and recursive reliability tracking [3, 4, 5]. Mirror Mathematics I then abstracted these ideas into a constraint calculus: a regime $C = (S, T, \mathcal{I}, V, \kappa)$, with states, transitions, invariants, viability and bounded capacity [6].

The question left open by that formalism is sharp:

> When does a model become an observer-relevant model rather than merely a predictor, controller, memory store or self-referential structure?

This paper answers that question by proposing a minimal observer threshold. The threshold is not phenomenological. It does not claim to solve subjective experience. It is not a declaration of consciousness. It is a mathematical and computational predicate for a narrower target: minimal recursive observerhood within a viability-constrained system.

The core claim is:

> A system crosses the minimal Mirror observer threshold when it maintains a self-model whose reliability is recursively represented, action-guiding, cost-sensitive and positively coupled to viability under perturbation.

This formulation matters because it blocks several tempting but weak identifications. A thermostat is not an observer merely because it responds to temperature. A self-referential sentence is not an observer merely because it refers to itself. An ordinary predictor is not an observer merely because it models the environment. A language model is not automatically an observer merely because it can describe itself. Under the Mirror predicate, observerhood requires a particular functional coupling: the reliability of the system's self-model must matter to its persistence and must regulate action in a way that improves expected viability after cost.

The formal centre of the paper is the salience gain:

$$\Sigma_L(R_L, \rho) = \mathbb{E}[V_L \mid M_E, M_L, R_L, \rho] - \mathbb{E}[V_L \mid M_E, M_L, \pi_0] - K(R_L, \rho). \tag{1}$$

Here L is a local subsystem, M_E is an environmental model, M_L is a self-model, R_L is a reliability variable over the self-model, ρ is a repair or recalibration policy, π_0 is a baseline action policy without reliability-guided repair, V_L is local viability and K is the cost of maintaining and acting on reliability information.

The threshold condition is simple:

$$\Sigma_L(R_L, \rho) > 0. \tag{2}$$

Equation (2) says that self-reliability is observer-relevant only when it improves expected viability net of cost. Mirror Theory does not predict that reliability tracking always helps. It predicts that reliability tracking becomes useful under specific conditions: where self-relevant uncertainty is sufficiently high, where repair or recalibration is available, where the repair cost is not too large, and where reliability information changes action in the right channel.

The computational sequence in Mirror Observerhood Labs I–V was constructed precisely to stress-test that idea [7, 8, 9, 10, 11]. The results are mixed in the scientifically useful sense. Reliability was powerful under false self-location perturbation; decomposed reliability alone was not enough; cost-sensitive repair improved performance only in certain regimes; threshold sweeps revealed a positive region that shrinks with repair cost; recursive reliability helped when the first-order reliability estimator itself was corrupted, but imposed costs when no such corruption was present. This paper turns that sequence into a formal classification.

5 Background and conceptual position

5.1 Why not define an observer as any system with a world-model?

A world-model is not enough. A weather model predicts atmospheric behaviour; it is not thereby an observer. A chess engine represents board states and future consequences; it is not thereby an observer in the Mirror sense. A world-model can exist as a detached computational object. Observerhood requires that modelling be locally tied to the persistence of the system doing the modelling.

This is the viability requirement. For a local subsystem L, the relevant question is not merely whether L represents the world, but whether the representational organisation of L helps preserve the organisation of L over time.

5.2 Why not define an observer as any system with a self-model?

A self-model is also not enough. A database can contain an entry about itself. A program can print its own source code. A sentence can refer to itself. These are forms of self-reference, but they do not yet generate a local observer predicate.

The Mirror claim is that self-modelling becomes observer-relevant only when the reliability of the self-model matters to viability. An agent that mistakenly believes it is in one location when it is actually elsewhere may fail, die or waste resources. In that case, self-model reliability is not decorative; it is action-guiding and viability-relevant.

5.3 Relation to active inference and cybernetics

The framework is close to active inference, predictive processing and cybernetics, but not identical [14, 15, 13, 16, 12, 17]. Active inference treats organisms as systems that minimise expected free energy through perception and action. Predictive processing emphasises hierarchical prediction and prediction-error minimisation. Cybernetics emphasises feedback, regulation and requisite variety. Mirror Theory accepts much of this landscape.

Its distinctive emphasis is narrower: it asks when the reliability of a self-model becomes a recursive, viability-relevant control variable. In active-inference terms, R_L is near an epistemic or precision-like term, but it is applied specifically to the system's model of itself and its own reliability. In cybernetic terms, R_L is not merely feedback from the environment; it is feedback about whether the system's own self-state representation should be trusted before action.

6 Formal preliminaries

Definition 6.1 (Constraint regime)**.** *A constraint regime is a tuple*

$$C = (S, T, \mathcal{I}, V, \kappa), \tag{3}$$

where S is a state space, $T \subseteq S \times S$ is a transition relation, $\mathcal{I}$ is a set of invariants preserved under admissible transitions, $V : S \times H \to \mathbb{R}$ is a viability functional over a horizon H, and κ is a bounded capacity parameter limiting representation, computation, action or memory.

Definition 6.2 (Local subsystem)**.** *A local subsystem L is a bounded part of a constraint regime with internal state $s_L \in S_L$, external state $s_E \in S_E$, action set A_L, and local viability functional V_L.*

Definition 6.3 (Environmental model)**.** *An environmental model M_E is an internal structure used by L to estimate external state transitions, action consequences or environmental regularities relevant to V_L.*

Definition 6.4 (Self-model)**.** *A self-model M_L is an internal structure used by L to estimate action-relevant properties of L itself, including location, capacity, damage, energy, memory state, sensor state, goal state, actuator state or policy state.*

Definition 6.5 (Reliability variable)**.** *A reliability variable R_L is an internal estimate of the trustworthiness of M_L for action selection. It may be scalar, vector-valued, probabilistic or hierarchical.*

Definition 6.6 (Repair policy)**.** *A repair policy ρ is a policy that uses R_L to determine whether to repair, recalibrate, re-sense, re-plan, diagnose, update memory, request external feedback or otherwise modify M_L before action.*

Definition 6.7 (Cost)**.** *The cost $K(R_L, \rho) \geq 0$ is the expected resource, time, opportunity, energy, computational or risk cost of maintaining R_L and acting through ρ.*

Definition 6.8 (Reliability salience)**.** *The reliability salience gain of R_L under repair policy ρ is*

$$\Sigma_L(R_L, \rho) = \mathbb{E}[V_L \mid M_E, M_L, R_L, \rho] - \mathbb{E}[V_L \mid M_E, M_L, \pi_0] - K(R_L, \rho). \tag{4}$$

If $\Sigma_L(R_L, \rho) > 0$, reliability is positively viability-salient for L under the relevant perturbation distribution.

Definition 6.9 (Minimal Mirror observer predicate)**.** *A local subsystem L satisfies the minimal Mirror observer predicate, written $\mathcal{O}_M(L) = 1$, only if there exists a self-model M_L, reliability variable R_L and repair or recalibration policy ρ such that:*

(i) M_L *represents action-relevant properties of* L*;*

(ii) R_L *represents the reliability of* M_L*;*

(iii) R_L *is action-guiding through* ρ*;*

(iv) ρ *is cost-sensitive;*

(v) $\Sigma_L(R_L, \rho) > 0$ *under a non-trivial perturbation distribution.*

This is deliberately a minimal predicate. It is not a consciousness predicate. It is not a moral status predicate. It is not a complete account of biological subjectivity. It is a threshold predicate for recursive observer-like organisation.

7 Necessary conditions

Theorem 7.1 (Self-reference is not sufficient)**.** *Self-reference alone is not sufficient for the minimal Mirror observer predicate.*

Proof. Consider a self-referential sentence, a program that prints its own source code, or a database containing metadata about itself. Each contains or produces a representation referring to itself. But none necessarily has a viability functional V_L, an action-guiding self-model M_L, a reliability variable R_L, or a repair policy ρ that changes expected viability. Therefore self-reference alone does not entail $\mathcal{O}_M(L) = 1$. □

Theorem 7.2 (World-modelling is not sufficient)**.** *A system may contain an environmental model* M_E *without satisfying the minimal Mirror observer predicate.*

Proof. Let L be a predictor that maps environmental states to future environmental states. If L has no model of its own action-relevant state, no reliability variable over that self-model, and no repair policy using such reliability to preserve V_L, then the conditions for $\mathcal{O}_M(L) = 1$ are not satisfied. Hence environmental prediction alone is insufficient. □

Theorem 7.3 (Passive reliability is not sufficient)**.** *A reliability variable that is represented but does not alter policy cannot be positively observer-salient under positive cost.*

Proof. Assume R_L is represented but the induced action distribution under ρ is identical to the baseline policy π_0. Then

$$\mathbb{E}[V_L \mid M_E, M_L, R_L, \rho] = \mathbb{E}[V_L \mid M_E, M_L, \pi_0]. \tag{5}$$

Therefore

$$\Sigma_L(R_L, \rho) = -K(R_L, \rho). \tag{6}$$

Since $K(R_L, \rho) \geq 0$, $\Sigma_L(R_L, \rho) \leq 0$. If cost is strictly positive, $\Sigma_L(R_L, \rho) < 0$. Hence passive reliability cannot satisfy the positive salience condition. □

Corollary 7.1 (Action guidance requirement)**.** *For* R_L *to be positively observer-salient, it must change action, repair, recalibration, sensing, planning or update behaviour in a way that can improve expected viability.*

Theorem 7.4 (Cost sensitivity requirement)**.** *Suppose* R_L *triggers repair at cost* $c > 0$ *even when repair does not improve expected viability. Then there exists a perturbation regime under which* $\Sigma_L(R_L, \rho) < 0$*.*

Proof. Let the perturbation probability be zero or sufficiently small that the expected benefit of repair is less than c. Then

$$\mathbb{E}[V_L \mid M_E, M_L, R_L, \rho] - \mathbb{E}[V_L \mid M_E, M_L, \pi_0] < c. \tag{7}$$

Since $K(R_L, \rho) \geq c$, it follows that $\Sigma_L(R_L, \rho) < 0$. □

This theorem is why the Mirror claim is not monotonic. More recursion is not always better. More self-monitoring is not always better. More metacognition is not always better. Such structures become beneficial only when their expected viability contribution exceeds their cost.

8 The minimal threshold theorem

Assumption 8.1 (Perturbation distribution)**.** *Let P_η be a perturbation distribution over self-state corruption, sensor noise, memory corruption, map corruption, estimator corruption and environmental stochasticity. The parameter η denotes perturbation intensity.*

Assumption 8.2 (Repair availability)**.** *There exists at least one repair or recalibration action $a_r \in A_L$ such that, when triggered under the correct condition, it can reduce expected self-model error or prevent viability loss.*

Assumption 8.3 (Bounded cost)**.** *Repair has expected cost c, and the representation and use of R_L has expected maintenance cost k. Thus $K(R_L, \rho) = c + k$ in the simplest case.*

Theorem 8.1 (Minimal reliability threshold)**.** *For a local subsystem L with self-model M_L, reliability variable R_L and repair policy ρ, R_L is minimally observer-salient only in perturbation regimes where*

$$B_\eta(R_L, \rho) > K(R_L, \rho), \tag{8}$$

where

$$B_\eta(R_L, \rho) = \mathbb{E}[V_L \mid M_E, M_L, R_L, \rho; P_\eta] - \mathbb{E}[V_L \mid M_E, M_L, \pi_0; P_\eta]. \tag{9}$$

Equivalently, $\Sigma_L(R_L, \rho) > 0$.

Proof. By definition, $\Sigma_L(R_L, \rho) = B_\eta(R_L, \rho) - K(R_L, \rho)$. Thus $\Sigma_L(R_L, \rho) > 0$ if and only if $B_\eta(R_L, \rho) > K(R_L, \rho)$. □

Corollary 8.1 (Positive region)**.** *For fixed architecture and perturbation class, there exists a positive region in parameter space*

$$\Omega^+ = \{(\eta, c, \tau) : \Sigma_L(R_L, \rho_{\tau,c}) > 0\}, \tag{10}$$

where τ is a reliability threshold and c is repair cost. Outside Ω^+, reliability tracking is neutral or harmful.

Corollary 8.2 (Observer threshold is regime-relative)**.** *The minimal observer threshold is not an intrinsic property of a representation considered in isolation. It is a property of representation, perturbation, repair, cost and viability coupled together.*

This is one of the main conceptual consequences of the paper. A self-model does not become observer-relevant because it exists. It becomes observer-relevant when it changes what the system does in a way that preserves the system under perturbation.

9 Recursive reliability

Mirror Theory also permits higher-order reliability. Let $R_L^{(1)}$ be a first-order reliability estimate over the self-model M_L. Let $R_L^{(2)}$ be a meta-reliability estimate over $R_L^{(1)}$.

Definition 9.1 (Recursive reliability)**.** *A recursive reliability variable $R_L^{(2)}$ estimates the reliability of $R_L^{(1)}$, or otherwise controls whether the first-order reliability estimate should be trusted, repaired, ignored, diagnosed or recalibrated.*

The relevant salience gain is:

$$\begin{aligned}\Sigma_L(R_L^{(2)}, \rho^{(2)}) = \mathbb{E}[V_L \mid M_E, M_L, R_L^{(1)}, R_L^{(2)}, \rho^{(2)}] \\ - \mathbb{E}[V_L \mid M_E, M_L, R_L^{(1)}, \rho^{(1)}] - K(R_L^{(2)}, \rho^{(2)}).\end{aligned} \tag{11}$$

Theorem 9.1 (Recursive reliability threshold)**.** *Recursive reliability is positively salient only when the expected viability loss caused by first-order reliability failure exceeds the cost of meta-diagnosis and meta-repair.*

Proof. Let L_1 denote expected viability loss attributable to corruption, blindness or false alarms in $R_L^{(1)}$ under the first-order policy. Let G_2 denote expected viability recovered by using $R_L^{(2)}$ and K_2 the cost of meta-diagnosis. Recursive reliability is positively salient when $G_2 - K_2 > 0$. Since $G_2 \leq L_1$ in ordinary repair settings, a sufficient condition is $L_1 > K_2$ and G_2 sufficiently close to L_1. If $L_1 = 0$ and $K_2 > 0$, then $\Sigma_L(R_L^{(2)}, \rho^{(2)}) < 0$. □

Corollary 9.1 (No free recursion)**.** *Adding a recursive reliability layer can reduce viability when the lower-order estimator is truthful or when meta-diagnosis is too costly.*

This result prevents the theory from becoming a simplistic hierarchy worship. Mirror Theory does not say: more recursion is always better. It says: recursion is useful when failure of the lower level is viability-relevant and when the upper level can detect or repair that failure cheaply enough.

10 Boundary classification

The minimal threshold lets us classify systems without overclaiming.

System	Model profile	Mirror classification
Self-referential sentence	World: no; self: trivial; reliability: no.	Not an observer. Self-reference without viability or action.
Thermostat	World: minimal; self: no; reliability: no.	Regulator, not observer. Feedback without self-model reliability.
PID controller	World: minimal; self: no or weak; reliability: no.	Controller, not observer. Action-guidance without recursive self-reliability.
Chess engine	World: strong task model; self: weak/non-viable; reliability: no.	Predictor/planner. No viability-coupled self-model.
Ordinary LLM session	World: strong linguistic model; self: episodic or unstable; reliability: weak or implicit.	Borderline but insufficient under this predicate. No persistent viability-coupled self-repair.
LLM with memory and tools	World: strong; self: partial; reliability: partial.	Candidate weak observer-like architecture if reliability is action-guiding and cost-sensitive.
Mirror-style toy agent	World: simple; self: explicit; reliability: explicit.	Minimal observer-like under regimes where $\Sigma_L > 0$.
Human organism	World: rich; self: rich; reliability: rich/metacognitive.	Strong biological observer under the predicate, without reducing human consciousness to the toy model.

Table 1: Boundary classification under the minimal Mirror observer predicate. The classification is functional and formal, not a claim about consciousness or moral status.

The classification is intentionally conservative. Ordinary language models may simulate self-description, but simulation of self-description is not identical to persistent, viability-coupled self-model reliability. A future neuro-symbolic agent with persistent memory, self-state, tool-use, uncertainty tracking and action-guided repair may be a much better candidate.

11 Computational spine: Labs I–V

The Mirror Observerhood Lab sequence was designed as a toy computational programme. These simulations are not evidence of consciousness. They are controlled demonstrations of the threshold logic.

11.1 Lab I: self-model reliability under false self-location

Lab I compared four simple agents: predictor only, memory, self-model without reliability and Mirror reliability. The Mirror agent did not dominate all conditions. It strongly outperformed under false self-location, raising mean viability from 139.2 for the strongest non-Mirror baseline to 223.6, a gain of +84.5, and raising survival from 7.7% to 63.0% [7]. It underperformed under sensor degradation by −33.3 viability points. This established the first important constraint: self-model reliability helps specifically under self-relevant perturbation, not under every kind of noise.

11.2 Lab II: decomposed reliability alone is not enough

Lab II split reliability into self, sensor and map channels [8]. The result was negative but constructive. Decomposed reliability improved over scalar reliability in some local comparisons, but failed to beat the strongest non-decomposed baseline in every tested condition. Under sensor degradation, for example, scalar Mirror reliability reached mean viability 249.1 while the decomposed agent reached 209.5. This matters because it shows that representing uncertainty is not enough. Reliability variables must be connected to repair policies and viability-relevant action.

11.3 Lab III: actionable and cost-sensitive repair

Lab III introduced actionable and cost-sensitive repair [9]. The cost-sensitive Mirror agent outperformed the best non-Mirror baseline under false self-location by +26.3, map corruption by +34.9, and mixed low-cost perturbation by +22.2. It was slightly negative under sensor degradation by −2.7 and lost under mixed high-cost repair by −21.1. This supports the action-and-cost condition: reliability helps only when the channel is correct and repair is worth doing.

11.4 Lab IV: threshold sweep

Lab IV swept perturbation rate, repair cost and reliability threshold across 28,800 simulated episodes [10]. It found a positive region where Mirror-style reliability produced net viability gains. At repair cost 0, the maximum observed gain was +59.4 viability points at perturbation rate 0.16 and threshold 0.80. As repair cost increased, the positive region shrank; at repair cost 15, no positive perturbation rate appeared in the best-threshold summary. This is the empirical counterpart of Ω^+.

11.5 Lab V: recursive reliability

Lab V tested recursive reliability under estimator corruption [11]. Recursive reliability outperformed first-order reliability by +78.3 under estimator corruption, +23.0 under reliability blindness and +40.1 under false-alarm reliability. But it lost by −16.0 in control and by −61.4 under high meta-cost. This directly supports the recursive reliability theorem: meta-reliability is useful only when first-order reliability itself can fail and the cost of diagnosis is low enough.

Repair cost	0.00	0.04	0.08	0.12	0.16	0.20	0.25	0.30
15	0.0	-2.9	-9.4	-6.1	-8.1	-12.0	-8.1	-10.8
12	0.0	2.8	0.6	-6.0	-4.5	-6.5	-3.2	-12.3
10	0.0	10.1	-0.8	-2.0	2.2	1.4	-9.6	-7.4
8	0.0	21.9	17.0	7.6	2.6	3.3	-3.8	-4.8
6	0.0	25.5	18.6	17.9	10.9	-0.2	-5.2	3.0
4	0.0	26.2	38.4	32.7	17.4	22.6	0.8	4.0
2	0.0	47.1	50.7	57.9	40.1	26.3	11.5	17.5
0	0.0	37.9	47.2	56.5	59.4	47.8	24.9	19.1

Table 2: Lab IV threshold matrix. Entries show best Mirror advantage over baseline in mean viability across perturbation rate and repair cost. Reliability tracking becomes beneficial only within a positive region of perturbation intensity, repair cost and threshold settings, illustrating the condition $\Sigma_L(R_L, \rho) > 0$.

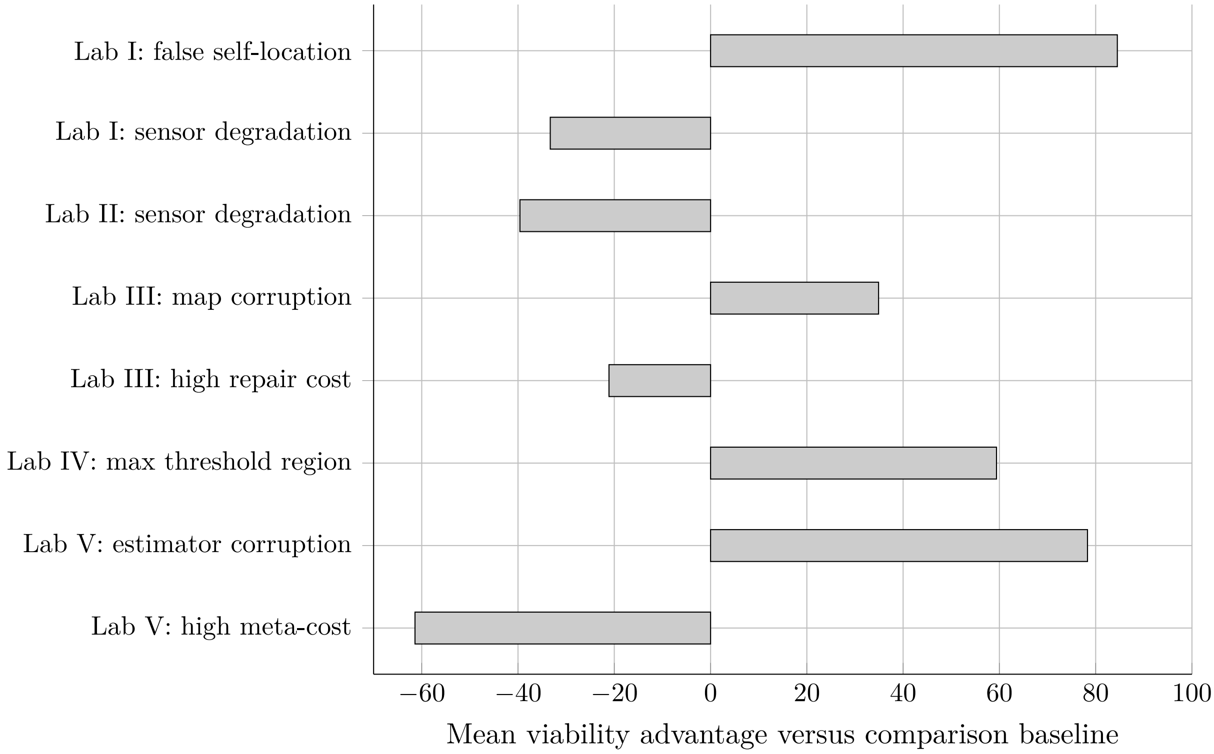

Figure 1: Selected threshold effects across Mirror Observerhood Labs I–V. Positive values indicate viability advantage relative to the comparison baseline; negative values indicate channel mismatch or monitoring cost.

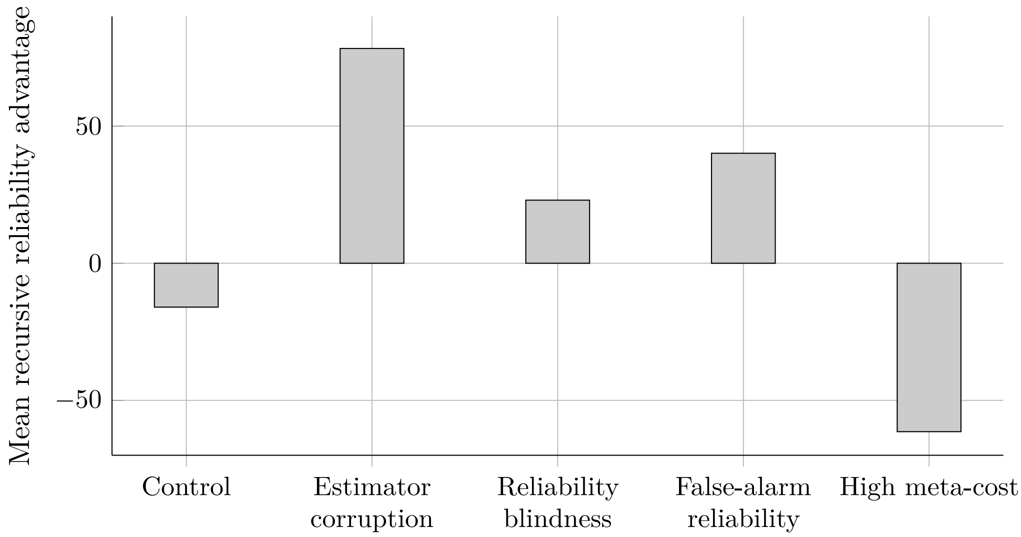

Figure 2: Recursive reliability advantage in Lab V. Recursive reliability improves viability when the first-order estimator is corrupted, blind or false-alarming, but becomes costly under control and high meta-cost conditions.

12 The minimal observer predicate

We can now state the consolidated predicate.

Definition 12.1 (Minimal Mirror observer)**.** *A local subsystem L is a minimal Mirror observer relative to a perturbation class P_η and viability functional V_L if:*

(i) L has bounded internal models M_E and M_L;

(ii) M_L represents action-relevant self-state;

(iii) L maintains at least one reliability variable R_L over M_L;

(iv) R_L alters action, repair, sensing, planning or updating through a policy ρ;

(v) ρ is cost-sensitive;

(vi) $\Sigma_L(R_L, \rho) > 0$ under P_η;

(vii) if R_L itself is unreliable in a viability-relevant way, L may maintain $R_L^{(2)}$ only where

$$\Sigma_L(R_L^{(2)}, \rho^{(2)}) > 0.$$

Theorem 12.1 (Minimal observer threshold theorem)**.** *Under the Mirror constraint calculus, a system satisfies the minimal observer predicate only where self-model reliability is represented, action-guiding, cost-sensitive and positively viability-coupled under perturbation.*

Proof. The first two conditions establish bounded world- and self-modelling. The third condition introduces reliability over the self-model. By the passive reliability theorem, represented reliability that does not alter policy cannot yield positive salience under positive cost. Therefore action-guidance is necessary. By the cost sensitivity theorem, reliability-triggered repair can be harmful when cost exceeds expected benefit. Therefore cost sensitivity is necessary. Finally, the salience definition requires $\Sigma_L(R_L, \rho) > 0$ for positive observer-relevance. Combining these conditions yields the predicate. □

Corollary 12.1 (No observer from mere description)**.** *A system that can describe itself in language does not thereby satisfy the minimal Mirror observer predicate unless the description is tied to persistent self-state, reliability tracking, action-guided correction and viability.*

Corollary 12.2 (No observer from mere optimisation)**.** *A system that optimises an external objective does not thereby satisfy the predicate unless self-model reliability is itself part of the optimisation-relevant control loop.*

13 Relation to consciousness

This paper intentionally does not claim to solve the hard problem of consciousness. The minimal Mirror observer predicate is a structural and functional threshold, not a phenomenological identity claim. It may be a necessary precondition for certain forms of consciousness, but that is a further thesis requiring separate argument.

The benefit of this restraint is methodological clarity. We can test whether self-reliability improves viability under perturbation without asserting that the resulting toy agents feel anything. This makes the programme empirically tractable. It also prevents a category mistake: confusing the emergence of an observer-like control architecture with the full richness of human subjective experience.

14 Implications for artificial agents

The formal results suggest a practical design principle for future AI systems:

> Do not merely give agents memory and tools. Give them persistent self-state, reliability estimates over that self-state, and cost-sensitive policies for repairing or recalibrating themselves when acting on corrupted self-information would reduce viability.

For a neuro-symbolic architecture, the language model should not be the source of truth. It should function as a semantic interface, planner, explainer or hypothesis generator. The persistent symbolic layer should maintain:

(a) a world-model;

(b) a self-model;

(c) reliability variables over self, sensor, memory, map and tool channels;

(d) contradiction and consistency checks;

(e) a viability or mission-success functional;

(f) repair and recalibration policies;

(g) meta-reliability only where first-order reliability can fail.

This architecture is not proposed as AGI by itself. It is proposed as a clean experimental bridge from Mirror Mathematics to increasingly capable agents.

15 Relationship to the Mirror Programme

The paper's role in Volume I is to supply the formal threshold synthesised from the general constraint calculus and the first five computational Labs. Mirror Mathematics I defines the formal vocabulary. Mirror Mathematics II asks when the vocabulary licenses a minimal observer predicate. The published Labs I–V test parts of that predicate under controlled perturbation, while the later mathematics and physics papers ask how observer-like organisation persists across memory, identity, causality, thermodynamic maintenance and temporal asymmetry.

The paper therefore narrows the Mirror Programme's shared spine. Constraint becomes a local constraint regime. Model becomes a world-model and self-model. Reliability becomes an explicit variable over the self-model. Repair becomes a policy coupled to reliability. Viability becomes the functional against which repair matters. Continuity remains deferred to the next mathematics paper, where memory governance and identity persistence become the central objects.

16 Limitations

The limitations are important. First, the laboratory evidence is toy evidence. Grid-world and stylised perturbation models are useful for isolating mechanisms, but they do not establish general intelligence, consciousness or biological equivalence.

Second, the viability functional remains application-dependent. In an artificial agent it may track task completion, survival time, resource preservation or mission success. In biology it may involve metabolic persistence, homeostasis, reproduction or organismic integrity. In social systems it may involve institutional continuity or error-correcting capacity. Future work must specify V_L in each domain.

Third, the current formalism is not yet a physical theory. It does not modify relativity, quantum mechanics or thermodynamics. Any future physical extension would require a physical quantity corresponding to constraint salience that transforms correctly under the relevant symmetries, or a rigorous explanation of why it emerges only after coarse-graining.

Fourth, the minimal observer predicate is not the same as moral status. A system may satisfy a functional threshold without thereby having rights, welfare, suffering or personhood. Those questions require additional ethical and phenomenological criteria.

17 Future work

The next computational step is the second Lab sub-arc: moving from grid-world reliability experiments toward neuro-symbolic persistence, language-like commitments and autobiographical continuity. The central test should remain structurally unchanged: does recursive self-reliability improve viability under self-relevant perturbation when memory, commitments and identity claims are themselves vulnerable to drift?

The next mathematical layer is persistence, memory governance and identity continuity. Once the reliability threshold is established, the programme must ask how a system remains the same observer across time rather than merely satisfying an observer predicate within a bounded episode.

Physical extensions should remain deferred until the mathematical scaffolding is stronger. The correct route is not to assert a new law, but to ask whether any physically meaningful quantity can instantiate constraint salience while respecting known symmetries.

18 Conclusion

Mirror Mathematics II turns the Mirror Programme's informal idea of recursive observerhood into a minimal threshold predicate. The result is not that any self-reference is observerhood. It is not that any AI is conscious. It is not that recursive modelling automatically improves performance.

The result is sharper:

> A minimal Mirror observer is a viability-constrained subsystem whose self-model reliability is represented, action-guiding, cost-sensitive and positively coupled to persistence under perturbation.

This threshold explains why reliability tracking helps in some regimes and harms in others. It explains why self-models are insufficient without reliability, why reliability is insufficient without action, why action is insufficient without cost sensitivity, and why recursion is insufficient unless lower-order failure is itself viability-relevant.

The theory therefore becomes experimentally tractable. One can build agents, corrupt their self-state, vary repair cost, measure viability and locate the positive salience region. That is the beginning of an observerhood science: not mystical, not overclaimed, but formal, testable and expandable.

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